Practical 05: Fragment-Based UI Navigation

Source: Android Programming Practical(1).pdf | Source pages: 37–45

Aim

To develop and test an Android application demonstrating fragment-based ui navigation.

Steps to Perform the Practical

1. Create a new Android Studio project for fragment navigation.

2. Create the fragments and layout XML files shown in the source practical.

3. Add the navigation buttons or controls to the main activity layout.

4. Use FragmentManager and FragmentTransaction to replace or show the required fragment when a control is selected.

5. Run the app and tap each navigation control.

6. Verify that the correct fragment screen appears for each selection and use Back if the source code adds transactions to the back stack.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

5. Develop an Android application implementing Fragment-based UI navigation between multiple screens.

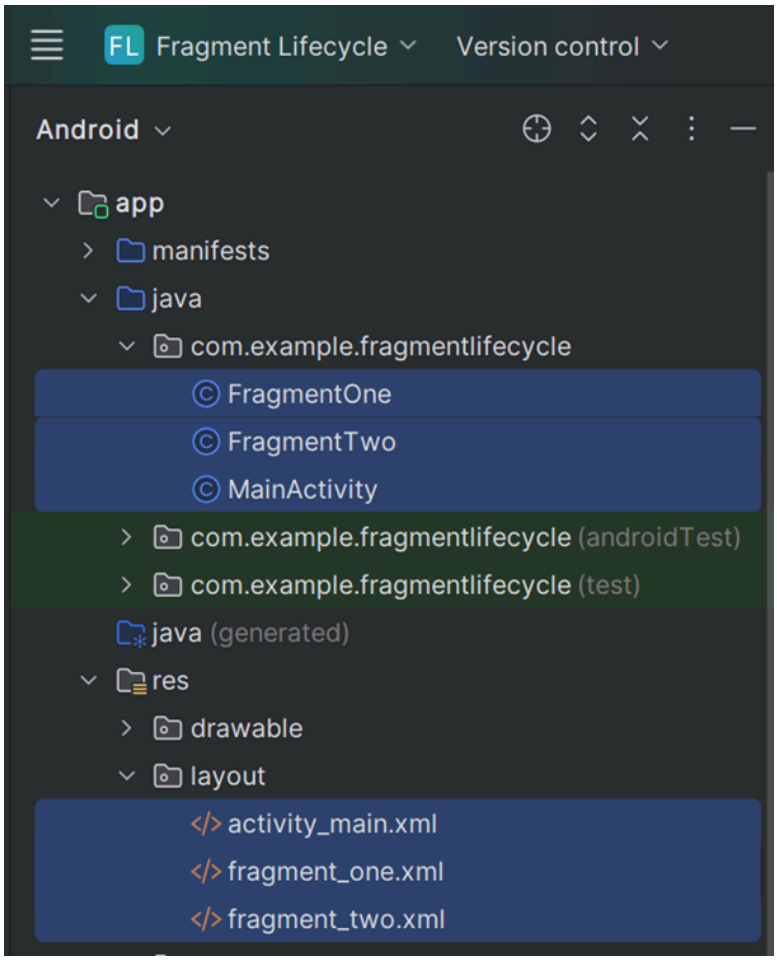
Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

Project Name: Fragment Lifecycle

Template: Empty Views Activity

Language: Java

Project Architecture: Total 6 files



I Filename: MainActivity.java

```
package com.example.fragmentlifecycle;

 import androidx.appcompat.app.AppCompatActivity;
 import android.os.Bundle;
 import android.widget.Button;

 public class MainActivity extends AppCompatActivity {

      private boolean isFragmentOneLoaded = true;
      private final androidx.fragment.app.FragmentManager fragmentManager =
getSupportFragmentManager();

      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

        Button btnChange = findViewById(R.id.btn_change);  // Ensure ID
matches activity_main.xml
      loadFragmentOne();

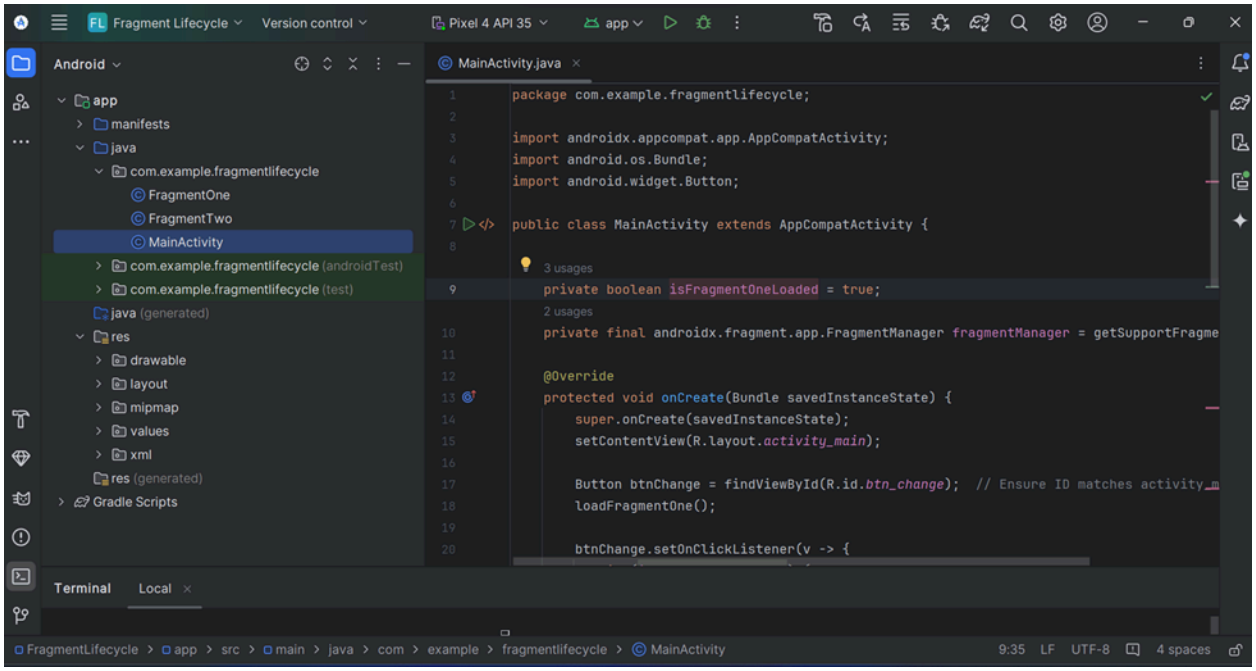
      btnChange.setOnClickListener(v -> {
            if (isFragmentOneLoaded) {
                  loadFragmentTwo();
            } else {
                  loadFragmentOne();
            }
      });
      }

      private void loadFragmentOne() {
         androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
      FragmentOne fragment = new FragmentOne();
      transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID
matches activity_main.xml
      transaction.addToBackStack(null);
      transaction.commit();
      isFragmentOneLoaded = true;
      }

      private void loadFragmentTwo() {
         androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
      FragmentTwo fragment = new FragmentTwo();
      transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID
matches activity_main.xml
```

```
        transaction.addToBackStack(null);
        transaction.commit();
        isFragmentOneLoaded = false;
        }
}
```

Path: app>java>com.example.fragmentlifecycle>MainActivity.java



II

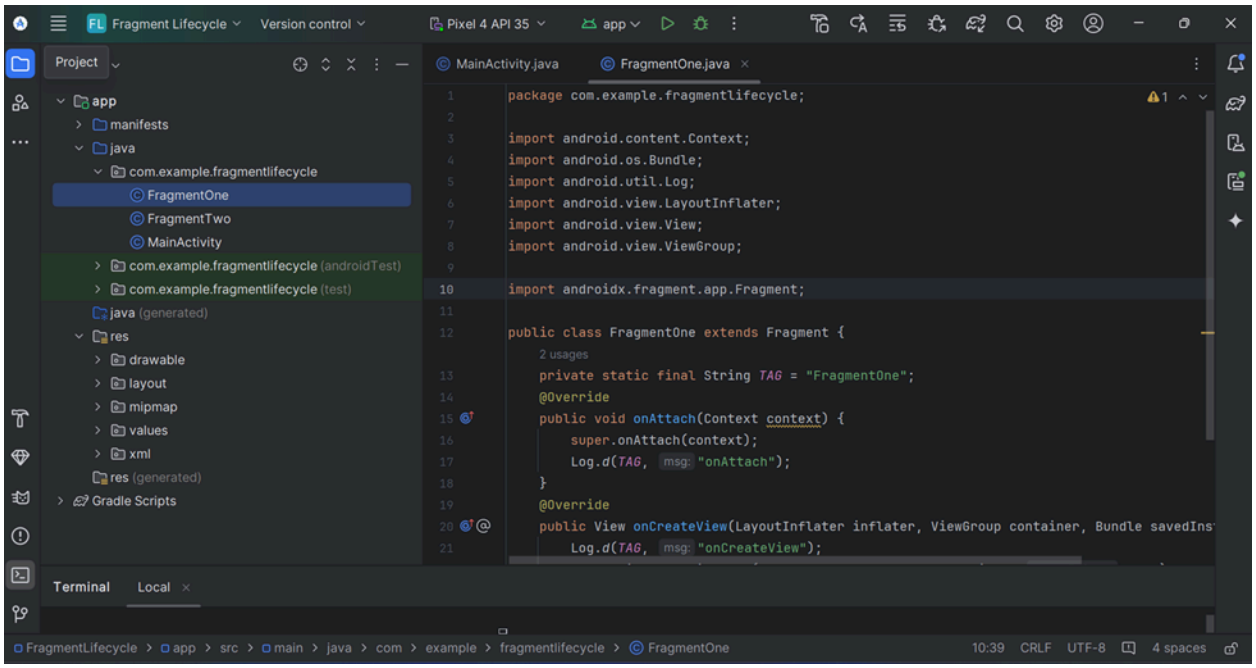
Filename: FragmentOne.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentOne extends Fragment {
        private static final String TAG = "FragmentOne";
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
```

```
        Log.d(TAG, "onAttach");
        }
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_one, container, false);
        }

}
```

Path: app>java>com.example.fragmentlifecycle > FragmentOne.java



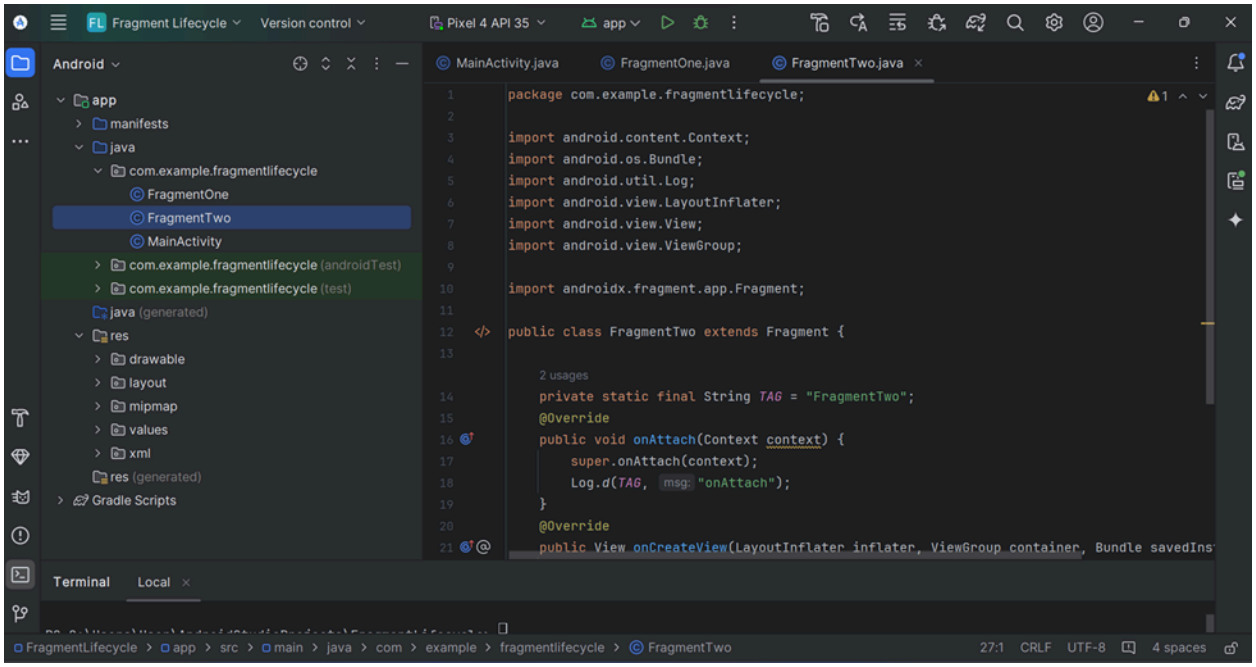
III

Filename: FragmentTwo.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentTwo extends Fragment {
        private static final String TAG = "FragmentTwo";
```

```
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
      Log.d(TAG, "onAttach");
        }\
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_two, container, false);
        }
}
```

Path: app>java>com.example.fragmentlifecycle > FragmentTwo.java



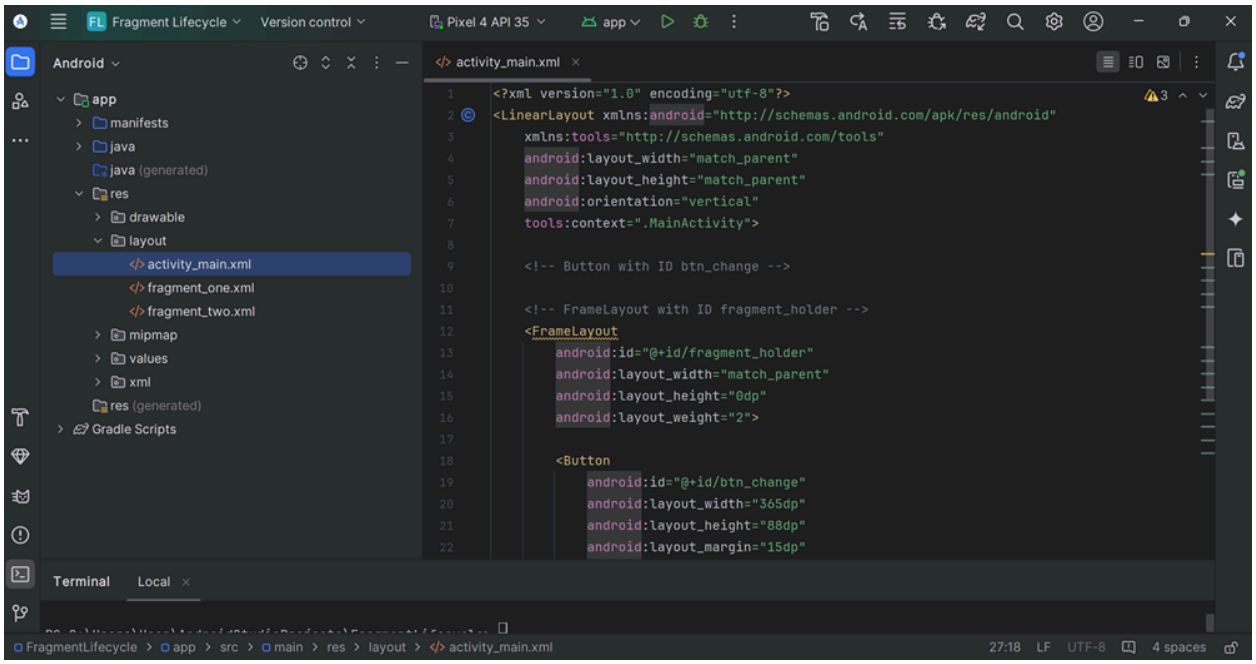
IV

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:tools="http://schemas.android.com/tools"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        tools:context=".MainActivity">
```

```
        <!-- Button with ID btn_change -->
        <!-- FrameLayout with ID fragment_holder -->
        <FrameLayout
        android:id="@+id/fragment_holder"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2">
        <Button
        android:id="@+id/btn_change"
        android:layout_width="365dp"
        android:layout_height="88dp"
        android:layout_margin="15dp"
        android:layout_marginTop="15dp"
        android:backgroundTint="@android:color/holo_green_dark"
        android:text="Change Fragment"
        android:textAllCaps="false"
        android:textSize="24sp" />
        </FrameLayout>
</LinearLayout>
```

Path: app>res>layout>activity_main.xml

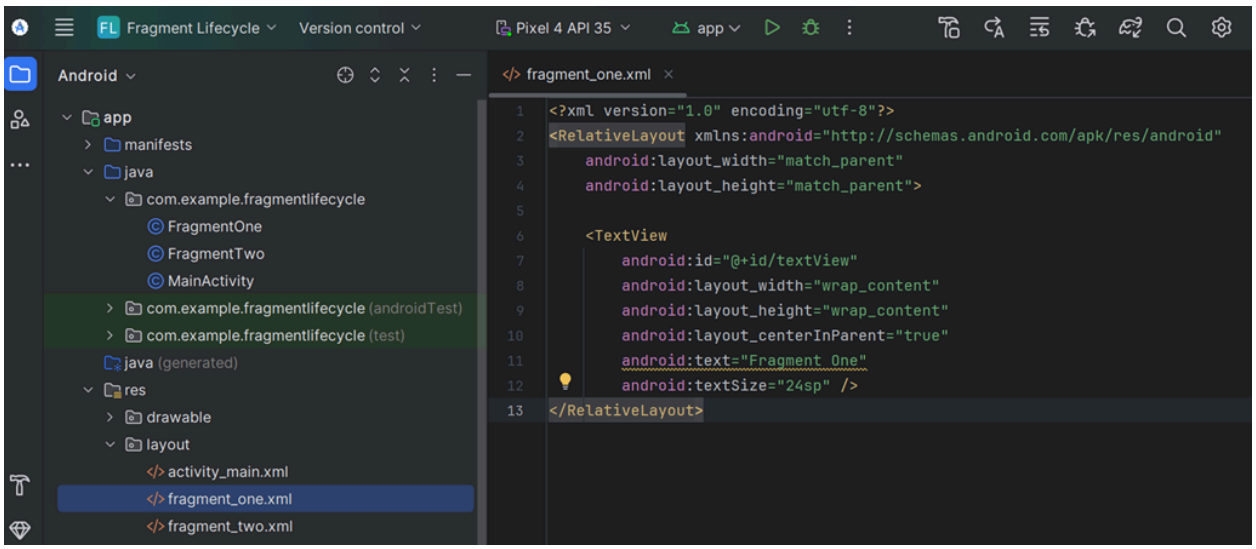


V

Filename: fragment_one.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment One"
        android:textSize="24sp" />
</RelativeLayout>
```

Path: app>res>layout>fragment_one.xml



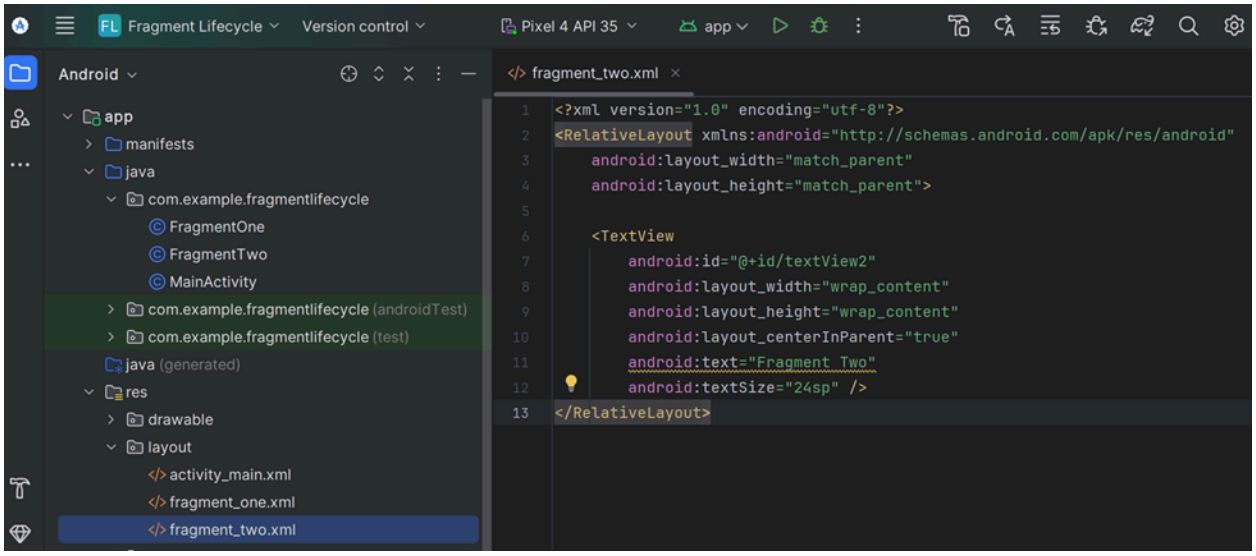
VI

Filename: fragment_two.xml

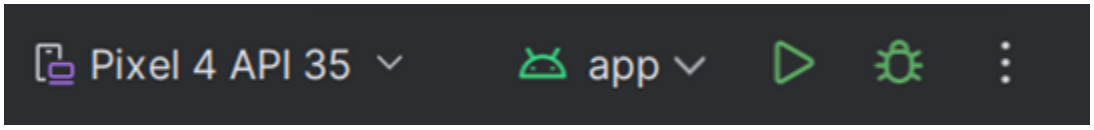
```
<?xml version="1.0" encoding="utf-8"?>
```

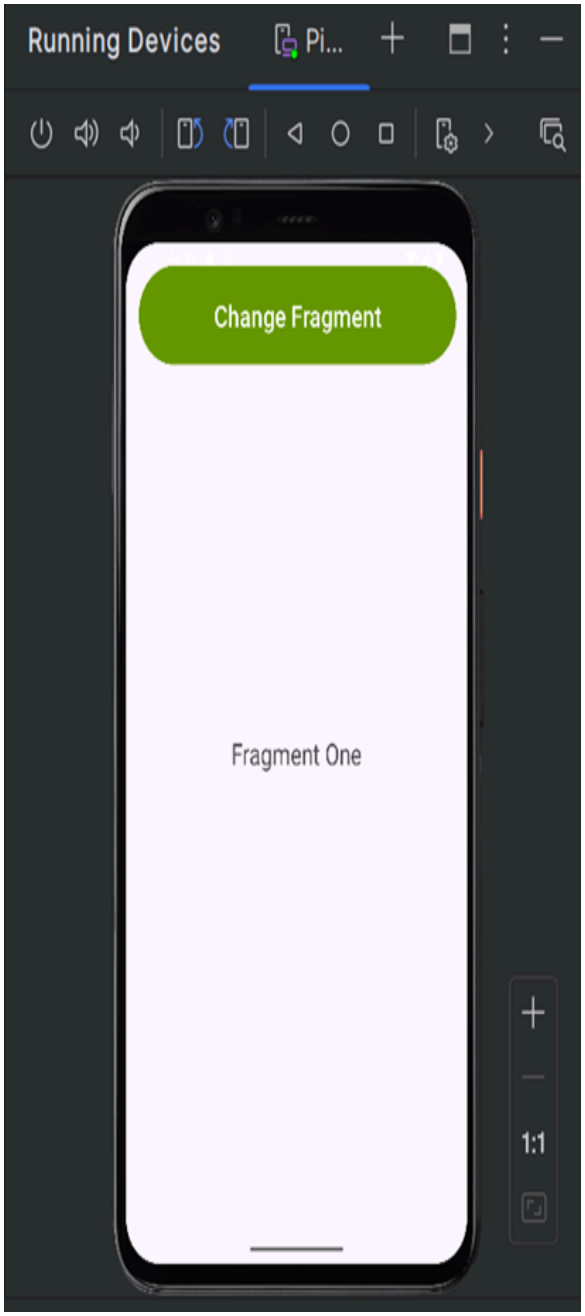
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment Two"
        android:textSize="24sp" />
</RelativeLayout>
```

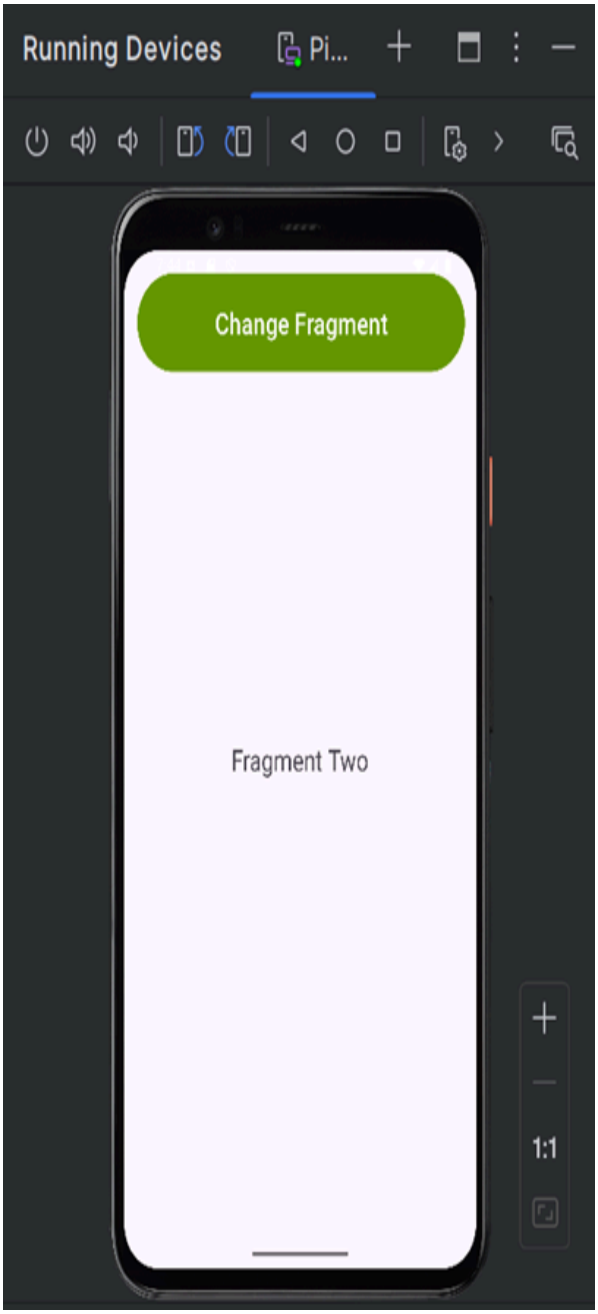
Path: app>res>layout>fragment_two.xml



Above the Android Studio near Main Menu Click the Play button to Run the Program




Running Devices
Pi...
Change Fragment
Fragment One
1:1


Running Devices
Pi...
Change Fragment
Fragment Two
1:1